

Why are paradigms not just systems of feature oppositions?

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1 Introduction

There is a broad consensus among morphologists that some form of paradigmatic organisation is found in inflection systems, capturing, for example, the fact that *buy* relates to *bought* in the same way as *stay* relates to *stayed*. A more contentious issue is whether this organisation can be reduced to a system of orthogonal contrasts in morphosyntactic feature values (see among many others Wunderlich & Fabri, 1995, p. 266). Recent work by Bonami et al. (2023) argues based on distributional data that it cannot: paradigm cells have properties that cannot be explained solely by their morphosyntactic features. They do this by showing that the distributional contrasts between pairs of cells that differ in the same features (e.g., NOM.SG ~ NOM.PL vs. ACC.SG ~ ACC.PL) are less parallel than one would expect under the usual assumptions.

Bonami et al. (2023) focused on Czech nouns and adjectives and did not attempt to understand *why* some contrasts in features led to a failure of parallelism. In this talk, we will take on this issue, looking into specific contrasts in Czech conjugation where a lack of distributional parallelism is expected on conceptual grounds.

Table 1 gives a snapshot of the relevant features of the Czech conjugation system. Perfective and imperfective verbs are distinct lexical items, even in cases like the one shown here, where a pair of verbs contrast only in aspect and not in lexical meaning. They also differ in the basic structure of their indicative subparadigms: perfective verbs lack present forms, and the vast majority of imperfectives express the future periphrastically. Past forms are synthetic only in the third person, where they are inflected for gender and number. Synthetic present and future forms are inflected for person and number.

Table 1: Active indicative synthetic forms of two Czech verbs differing in aspect.

	FUT	PRS	PST		FUT	PRS	PST
SG	1 <i>koupím</i>	—	<i>periphrastic</i>		<i>periphrastic</i>	<i>kupuji</i>	<i>periphrastic</i>
	2 <i>koupíš</i>	—	<i>periphrastic</i>		<i>periphrastic</i>	<i>kupuješ</i>	<i>periphrastic</i>
	3 <i>koupí</i>	—	<i>koupil(–/a/o)</i>		<i>periphrastic</i>	<i>kupuje</i>	<i>kupoval(–/a/o)</i>
PL	1 <i>koupíme</i>	—	<i>periphrastic</i>		<i>periphrastic</i>	<i>kupujeme</i>	<i>periphrastic</i>
	2 <i>koupíte</i>	—	<i>periphrastic</i>		<i>periphrastic</i>	<i>kupujete</i>	<i>periphrastic</i>
	3 <i>koupí</i>	—	<i>koupil(i/y/a)</i>		<i>periphrastic</i>	<i>kupují</i>	<i>kupoval(i/y/a)</i>
<i>koupit</i> ‘buy’ (perfective)				<i>kupovat</i> ‘buy’ (imperfective)			

Using this dataset, we can contrast two types of parallelism situations between contrasts. On the one hand, we expect to have strong parallelism when we compare the same pairs of contrasting person–number combinations across the two aspects, e.g. comparing PRS.1SG ~ PRS.1PL in imperfectives to FUT.1SG ~ FUT.1PL in perfectives, as there is no expectation that tense/aspect and person/number interact in complex ways. On the other hand, person and number interact in complex ways not only for verbs (see Siewierska, 2004; Corbett, 2000) but also, for example, pronouns (Harley & Ritter, 2002). As originally pointed out by Benveniste (1946), plurality does not have the same import in all persons, e.g., 1PL does not (typically) designate a plurality of speakers, while 3PL designates a plurality of entities that would be referred to individually in

the 3SG. Forchheimer (1953) discusses several conventional, language-specific uses of person–number combinations that do not match their ordinary semantics, such as politeness uses of 2PL to address a single individual or authorial use of 1PL (both are applied in Czech). This leads some authors (e.g., Wechsler, 2004) to reject the separation of person and number altogether. Without going that far, we have an expectation that contrasts that build on separating person and number, for instance, between PRS.1SG $\sim$ PRS.1PL and PRS.3SG $\sim$ PRS.3PL, will not be reliably parallel.

2 Methodology

Data We relied on the same data resources as Bonami et al. (2023), combining the inflectional morphological dictionary MorfFlexCZ 2.0 (Hajič et al., 2020), the SYN v9 corpus (Křen et al., 2021), and word vectors from the 100-dimensional vector space provided by Kyjánek & Bonami (2022).¹ We left out periphrastically expressed parts of the paradigm, and extracted pairs of future forms of perfective verbs (e.g., PFV.FUT.1SG $\sim$ PFV.FUT.1PL) as well as present tense forms of imperfectives (e.g., IPFV.PRS.1SG $\sim$ IPFV.PRS.1PL).

Method As illustrated in Figure 1, we followed the general methodology pioneered by Marelli & Baroni (2015): we trained a statistical model to predict word vectors in a target morphological category from those in a predictor category. In our case, the categories are cells in an inflectional paradigm. We then assessed the performance of the model by the cosine of the angle θ between the predicted vector $\vec{v}_{\text{predictor}}$ and the actual vector $(\vec{v}_{\text{actual}})$. Again, following Bonami et al. (2023), we used Gradient Boosting Tree regression models (Friedman, 2001) to predict each dimension in the target cell from all dimensions in the predictor cell. We report aggregated performance after 10-fold cross-validation.

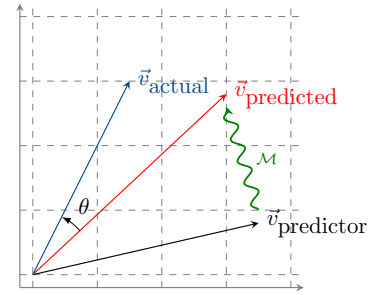


Figure 1: Vector prediction.

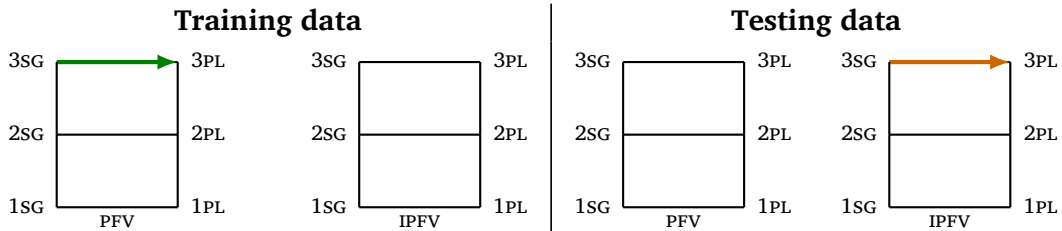


Table 2: Extrinsic prediction where number is parallel, person is constant but aspect is changed.

Predictions We evaluated the performance of our models in two ways, namely *intrinsic* and *extrinsic*, both sharing the same quality measure – the average cosine similarity between the predicted and actual vectors ($\cos(\vec{v}_{\text{predicted}}, \vec{v}_{\text{actual}})$). First, the intrinsic predictions assess discriminating power for predicting word vectors, i.e., the prediction of the same contrasts as the one on which the model was trained and, therefore, represents the traditional training-testing setup in machine learning. Second, the extrinsic predictions assess the stability of predicting word vectors in different contexts, i.e. the prediction of contrasts different from the one used

¹These models were trained as continuous bag of words (CBOW) models by applying word2vec (Mikolov et al., 2013) to the SYN v9 corpus. All relevant details on data preparation are described by Bonami et al. (2023).

for training the model. Table 2 illustrates such extrinsic prediction of the contrast specified as having a parallel number (SG $\sim$ PL) and constant person (3 $\sim$ 3) but being applied to a feature of a different aspect (PFV $\sim$ IPFV); as such, this exemplified how the prediction of PFV.FUT.3SG $\sim$ PFV.FUT.3PL can predict IPFV.PRS.3SG $\sim$ IPFV.PRS.3PL. We observed a total of ten types of predictions listed in Table 3; the boxes in the columns *Training* and *Testing* represent perfective and imperfective verbal paradigms on which the models were trained and tested; three horizontal lines in the boxes represent the person, while two vertical lines represent the number.

Table 3: Illustrations of predictions under consideration in Czech verbal subparadigms.

N.	Description of predictions	Training		Testing		# of datasets
		pfv	ipfv	pfv	ipfv	
(1)	Intrinsic					60
(2)	Parallel NUM, constant PER, across ASP					12
(3)	Parallel PER, constant NUM, across ASP					24
(4)	Parallel PER and NUM, across ASP					24
(5)	Parallel NUM across PER, constant ASP					24
(6)	Parallel NUM across PER and ASP					24
(7)	Parallel PER across NUM, constant ASP					24
(8)	Parallel PER across NUM and ASP					24
(9)	PER orthogonal to NUM, constant ASP					96
(10)	PER orthogonal to NUM across ASP					288

3 Results

Figure 2 summarizes the results. These are globally congruent with those reported in Bonami et al. (2023): (i) intrinsic predictions (row 1) are very accurate; (ii) orthogonal predictions (9–10) are very inaccurate on average, although some of the datasets lead to better results, probably by chance; (iii) overall, and parallel predictions (3–8) are slightly more accurate than orthogonal predictions, although there is a lot of overlap between the two.

The interesting and new result is the existence of differences among parallel cases. Two groups of situations are clearly distinguished. On the one hand, models trained on a particular person and/or number contrast and tested on the same contrast in a different tense/aspect configuration (2–4) perform quite well. On the other hand, models trained on a person (resp. number) contrast and tested in a different number (resp. person) perform markedly worse (5–8), doing barely better than orthogonal prediction. Thus, we take to be clear evidence confirming our expectation that person and number do not behave as orthogonal feature opposition, unlike person/number and tense/aspect.

In the talk, we will explore in more detail the differences between situations of parallel predictions and highlight the theoretical consequences of our findings.

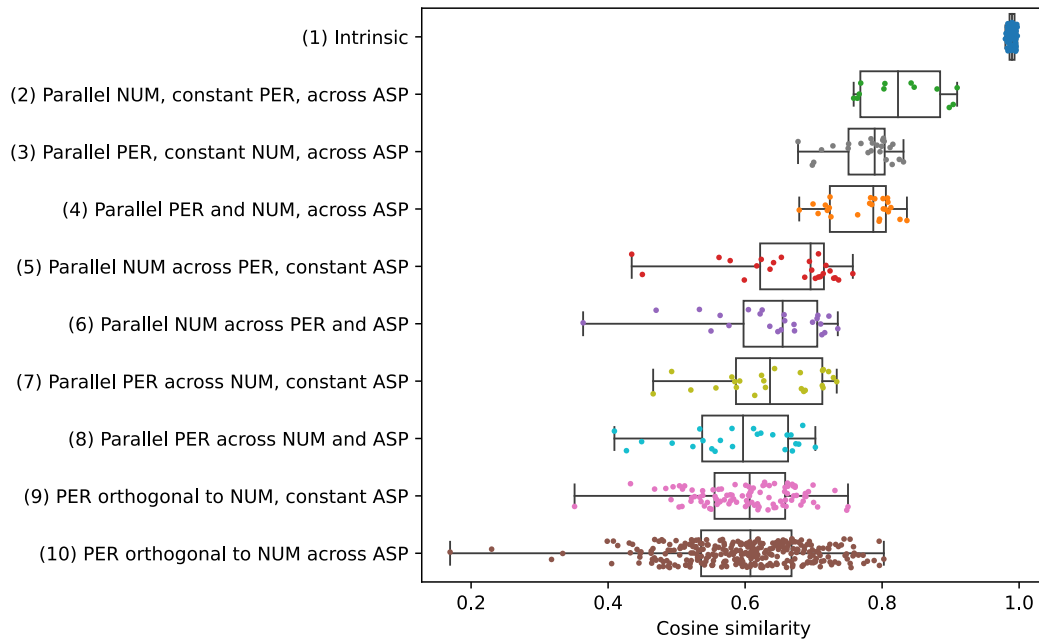


Figure 2: Results of intrinsic and extrinsic evaluation per individual types of predictions.

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